Statistical Analysis Project: COVID-19 Vaccination and Death Rates

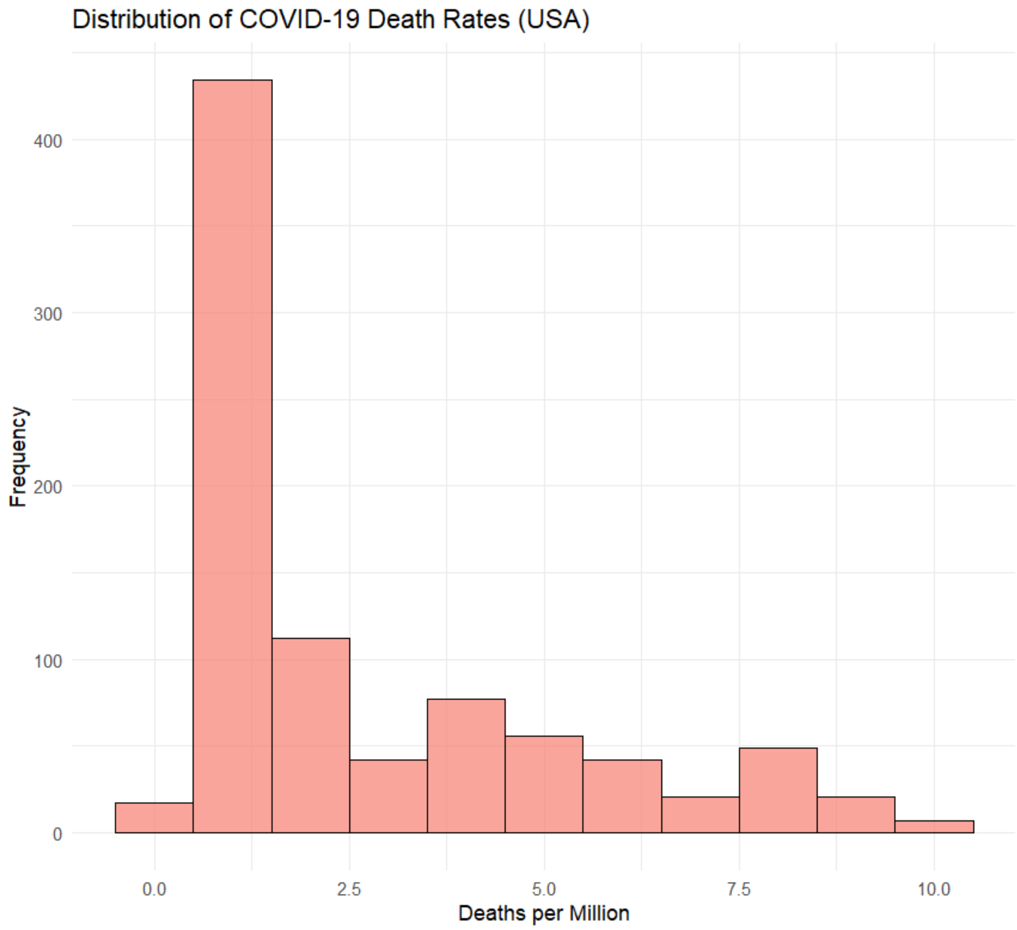
Introduction

This project analyzes the relationship between COVID-19 vaccination coverage and mortality rates in the United States. The dataset, sourced from *Our World in Data*, includes **878** daily observations ranging from late 2020 to 2023. The research question guiding this study is: **Is there a significant negative correlation between COVID-19 death rates and Vaccination rates?**

Part I: Central Tendency

We analyzed the center of our data distributions using Mean, Median, and Mode.

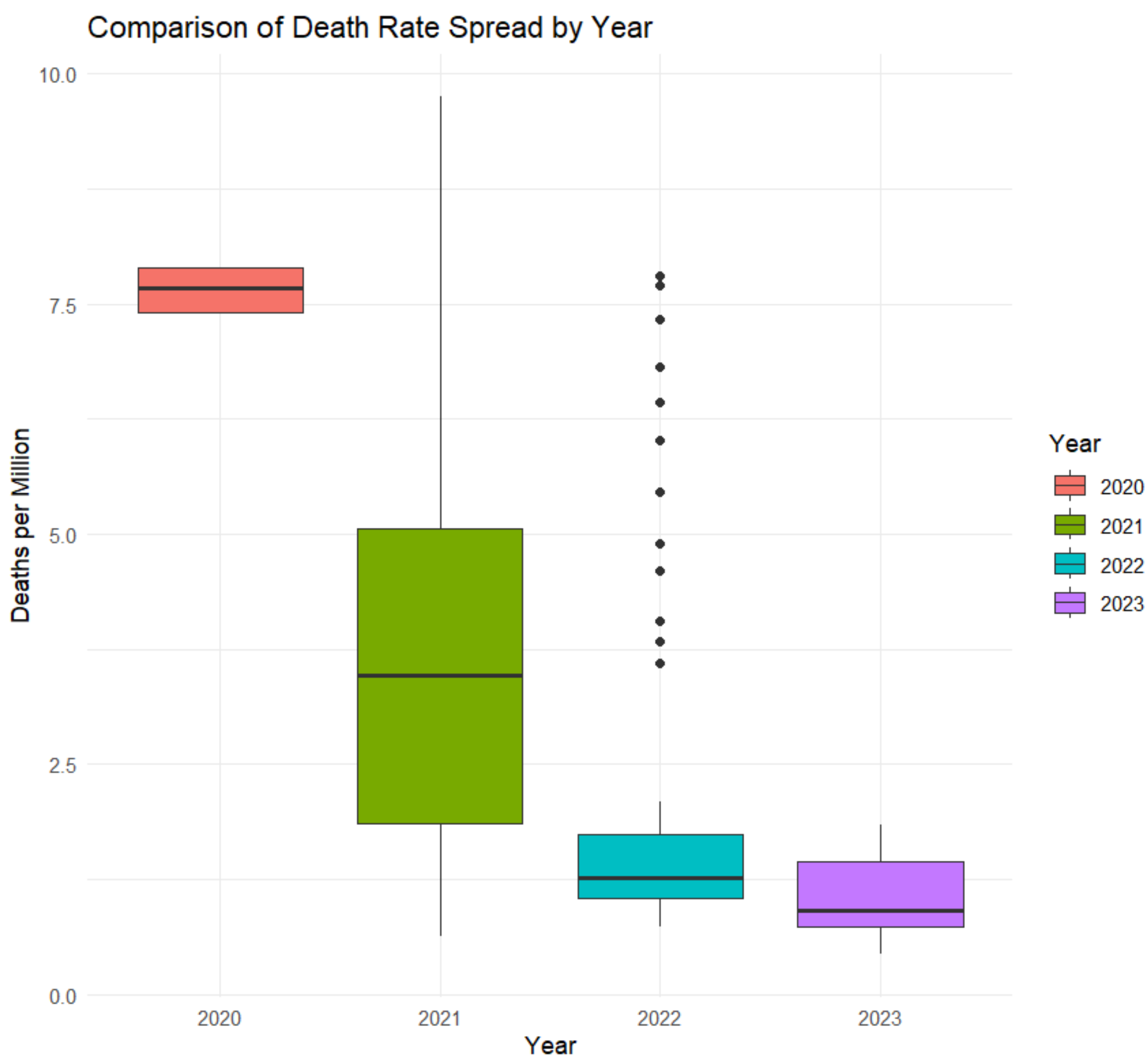
- **Vaccination Rate:** The average (mean) percentage of the population fully vaccinated during this period was **54.70%**, with a median of **65.69%**. The distribution is left-skewed, as indicated by the median being higher than the mean.
- **Death Rate:** The daily new confirmed deaths (per million) had a mean of **2.79**. The median was lower at **1.45**, suggesting that while typical days saw fewer deaths, there were periods of extreme spikes that pulled the average up.
- **Visualization:** The histogram of death rates reveals a **right-skewed distribution**, with a high frequency of days having low death rates (< 2 per million) and a long tail extending to over 9 deaths per million during peak waves.



Part II: Measures of Spread

To understand the variability in the data, we examined the Range, Interquartile Range (IQR), and Variance.

- **Vaccination Rate Variability:** The vaccination data covers the entire rollout period, resulting in a large spread. The variance is **439.73**. The values range from **0.00%** (start of rollout) to **69.47%** (maximum achieved). The Interquartile Range is **17.38**, representing the middle 50% of growth in vaccination coverage.
- **Death Rate Variability:** The death rate showed high variability with a variance of **5.99**. The values ranged from a minimum of **0.44** to a maximum of **9.75** deaths per million. The Interquartile Range was **3.20**.
- **Grouped Boxplot Analysis:** We created boxplots grouped by **Year** (2020-2023). This visualization clearly shows that the spread and median death rates were significantly higher in 2020 and 2021 (the height of the pandemic) compared to 2022 and 2023, where the boxes are much shorter and closer to zero, indicating reduced mortality and variability after widespread vaccination.
- **Outlier Analysis:** The boxplot and histogram reveal significant upper outliers. These "spikes" correspond to the peak winter wave in **January 2021**, where daily death rates repeatedly exceeded the upper fence (Q3 + 1.5*IQR). These extreme values pull the mean higher than the median, confirming the right-skewed shape described in Part I.

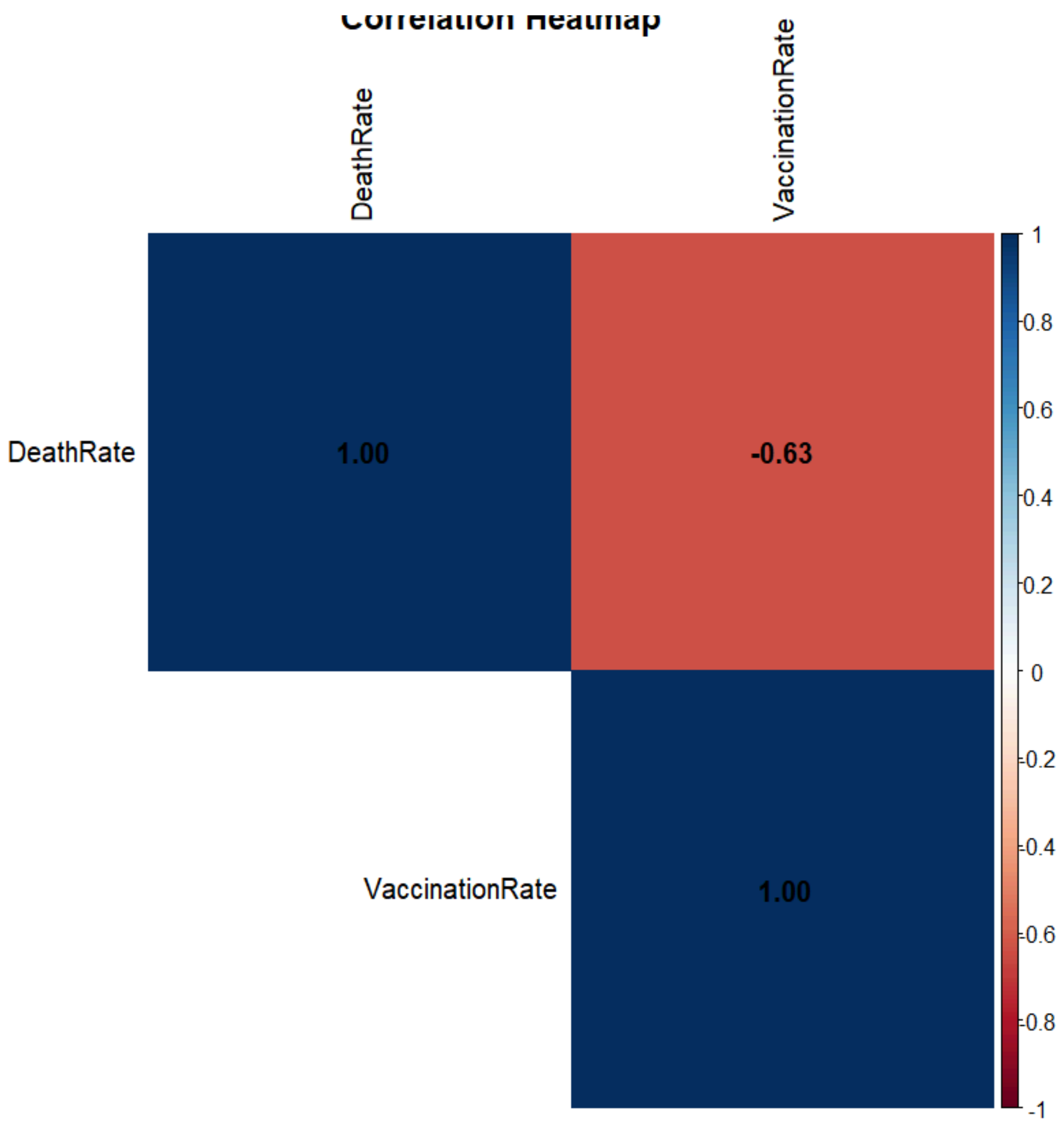


Part III: Correlation Analysis

We investigated the relationship between the two key variables:

1. **Independent Variable:** Vaccination Rate (People fully vaccinated per hundred).
2. **Dependent Variable:** Death Rate (New deaths smoothed per million).

- **Correlation Coefficient (r):** The calculated Pearson correlation is **-0.634**.
- **Interpretation:** This indicates a **strong negative correlation**. As the vaccination rate increases, the death rate tends to decrease.
- **Heatmap:** The correlation matrix heatmap confirms this inverse relationship, displaying a strong negative color intensity between the two variables.



Part IV: Simple Linear Regression

We fitted a linear regression model to quantify the relationship.

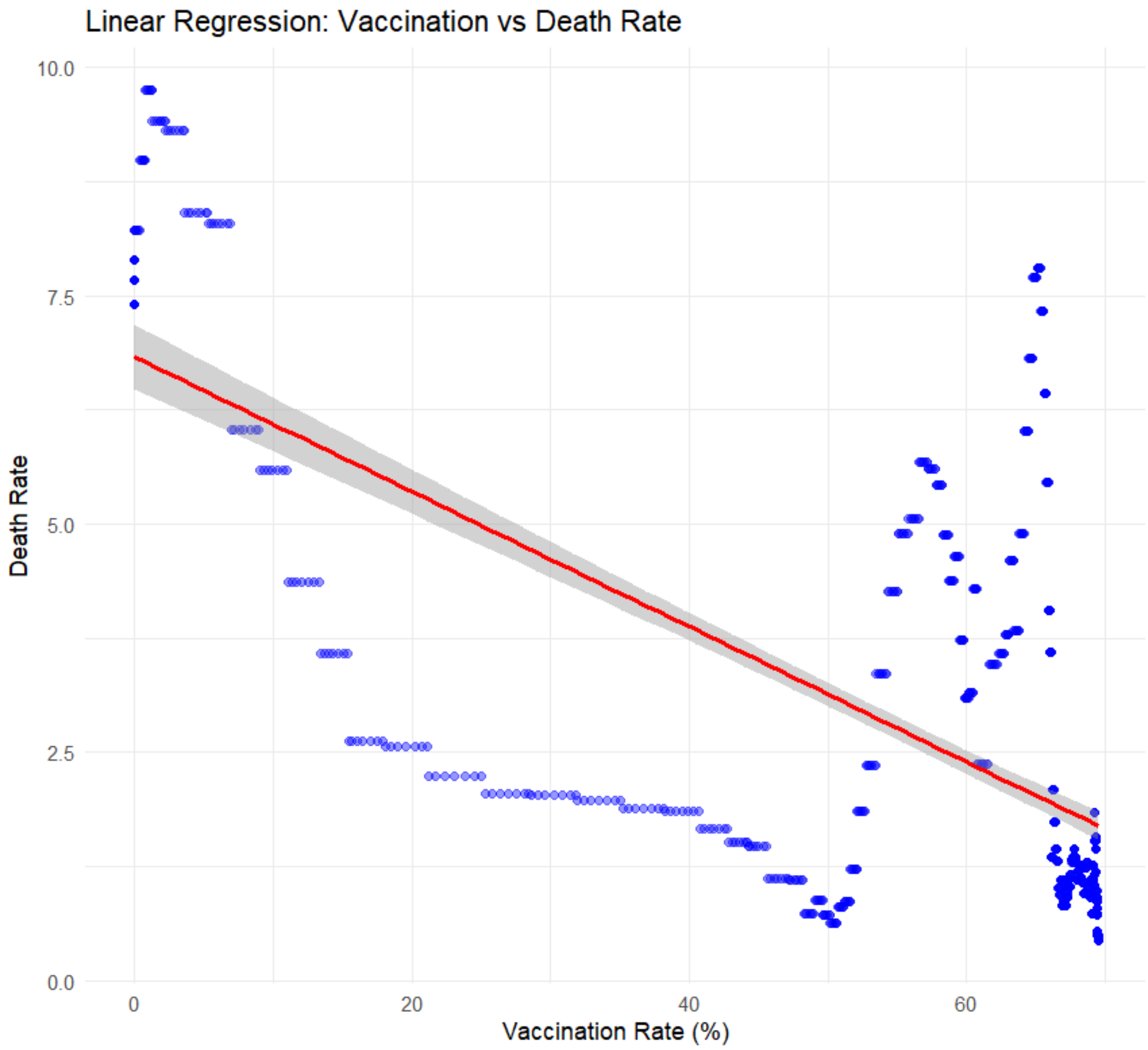
Regression Equation: y = -0.074x + 6.835

- **Slope (-0.074):** For every 1% increase in the fully vaccinated population, the daily death rate decreases by approximately **0.074** deaths per million.
- **Intercept (6.835):** The model predicts a theoretical death rate of 6.835 per million if the vaccination rate were 0%.

Prediction: Using our model, we predicted the death rate for a vaccination rate of **60%**.

- y = -0.074(60) + 6.835 = 2.40
- The model predicts approximately **2.40** deaths per million when 60% of the population is vaccinated.

Scatterplot: The scatterplot with the regression line overlaid shows the data points clustering downwards as vaccination rates move to the right, visually confirming the negative slope.



Part V: Statistical Inference

Finally, we estimated the population mean for the daily death rate using a t-test.

- **Sampling Distribution:** To estimate the population mean, we relied on the **Central Limit Theorem.** Since our sample size (n=878) is significantly larger than 30, we assume the sampling distribution of the sample mean is approximately normal. We utilized the t-distribution (with df = n-1) rather than the standard normal distribution because the population standard deviation is unknown.

Conclusion

The statistical analysis confirms a significant negative association between vaccination rates and COVID-19 death rates in the United States. The strong negative correlation (r = -0.63) and the negative slope in our regression model support the hypothesis that increased vaccination coverage contributed to lower mortality rates over the observed period.

R CODE

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# PROJECT: COVID-19 Vaccination Rates and Death Rates (Refined)
# DATASET: owid-covid-data.csv (Filtered for United States)
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# 0. SETUP AND LIBRARIES
# ---------------------------------------------------------------------------
# Ensure packages are installed: install.packages(c("dplyr", "ggplot2", "corrplot",
"modeest"))
library(dplyr)
library(ggplot2)
library(corrplot)
library(modeest) # For mode calculation

# Load Data
raw_data <- read.csv("owid-covid-data.csv")

# PREPROCESSING: Filter for USA and Select Variables
# We create a 'Year' column to satisfy the "Grouped Boxplot" requirement.
project_data <- raw_data %>%
  filter(location == "United States") %>%
  select(date, new_deaths_smoothed_per_million, people_fully_vaccinated_per_hundred)
%>%
  rename(Date = date,
        DeathRate = new_deaths_smoothed_per_million,
        VaccinationRate = people_fully_vaccinated_per_hundred) %>%
  na.omit() %>%
  mutate(Date = as.Date(Date),
        Year = as.factor(format(Date, "%Y"))) # Create categorical variable

head(project_data)

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# PART I: Exploratory Data Analysis - Central Tendency
# (Mean, Median, Mode)
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cat("\n--- PART I: MEASURES OF CENTRAL TENDENCY ---\n")

# Vaccination Rate
mean_vac <- mean(project_data$VaccinationRate)
median_vac <- median(project_data$VaccinationRate)
mode_vac <- mlv(project_data$VaccinationRate, method = "mfv")[1] # Most frequent value

cat(sprintf("Vaccination Rate -> Mean: %.2f, Median: %.2f, Mode: %.2f\n", mean_vac,
median_vac, mode_vac))

# Death Rate
mean_death <- mean(project_data$DeathRate)
median_death <- median(project_data$DeathRate)
mode_death <- mlv(round(project_data$DeathRate, 1), method = "mfv")[1] # Rounded
mode

cat(sprintf("Death Rate      -> Mean: %.2f, Median: %.2f, Mode: %.2f\n", mean_death,
median_death, mode_death))

# Visualization: Histogram (Distribution)
ggplot(project_data, aes(x=DeathRate)) +
  geom_histogram(binwidth=1, fill="salmon", color="black", alpha=0.7) +
  labs(title="Distribution of COVID-19 Death Rates (USA)", x="Deaths per Million",
y="Frequency") +
  theme_minimal()

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# PART II: Exploratory Data Analysis - Measures of Spread
# (Range, IQR, Variance)
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cat("\n--- PART II: MEASURES OF SPREAD ---\n")


# 1. Vaccination Rate Spread
range_vac <- range(project_data$VaccinationRate)
iqr_vac <- IQR(project_data$VaccinationRate)
var_vac <- var(project_data$VaccinationRate)

cat(sprintf("Vaccination Rate -> Range: [%.2f - %.2f], IQR: %.2f, Variance: %.2f\n",
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        range_vac[1], range_vac[2], iqr_vac, var_vac))

# 2. Death Rate Spread
range_death <- range(project_data$DeathRate)
iqr_death <- IQR(project_data$DeathRate)
var_death <- var(project_data$DeathRate)

cat(sprintf("Death Rate      -> Range: [%.2f - %.2f], IQR: %.2f, Variance: %.2f\n",
        range_death[1], range_death[2], iqr_death, var_death))

# --- Outlier Detection ---
# Identify specific outlier dates to support the report's conclusion
outlier_threshold <- quantile(project_data$DeathRate, 0.75) + 1.5 *
IQR(project_data$DeathRate)
outliers <- project_data %>%
  filter(DeathRate > outlier_threshold) %>%
  arrange(desc(DeathRate)) %>%
  select(Date, DeathRate)

cat("\n--- OUTLIER ANALYSIS ---\n")
cat("Upper Outlier Threshold (Q3 + 1.5*IQR):", round(outlier_threshold, 2), "\n")
cat("Top 5 Extreme Outlier Dates:\n")
print(head(outliers, 5))
# This proves the "January 2021" statement in the text

# Visualization: Grouped Boxplot (Comparing Spread by Year)
ggplot(project_data, aes(x=Year, y=DeathRate, fill=Year)) +
  geom_boxplot() +
  labs(title="Comparison of Death Rate Spread by Year", x="Year", y="Deaths per Million") +
  theme_minimal()
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# PART III: Correlation Analysis
# (Correlation Coefficient, Heatmap)
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cat("\n--- PART III: CORRELATION ANALYSIS ---\n")

# Calculate Pearson Correlation
correlation <- cor(project_data$VaccinationRate, project_data$DeathRate)
cat("Correlation Coefficient (r):", correlation, "\n")

# Visualization: Correlation Matrix Heatmap
# We select only numeric columns for the heatmap
numeric_data <- project_data %>% select(DeathRate, VaccinationRate)
cor_matrix <- cor(numeric_data)
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corrplot(cor_matrix, method="color", type="upper", addCoef.col = "black", tl.col="black",
title="Correlation Heatmap")

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# PART IV: Simple Linear Regression
# (Model, Prediction)
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cat("\n--- PART IV: LINEAR REGRESSION ---\n")

# Fit the model: DeathRate depends on VaccinationRate
model <- lm(DeathRate ~ VaccinationRate, data=project_data)
summary(model)

# Extract Coefficients
intercept <- coef(model)[1]
slope <- coef(model)[2]
cat(sprintf("Regression Equation: y = %.3fx + %.3f\n", slope, intercept))

# Prediction: Predict Death Rate if Vaccination Rate is 60%
new_data <- data.frame(VaccinationRate = 60)
prediction <- predict(model, newdata = new_data)
cat("Predicted Death Rate at 60% Vaccination:", prediction, "\n")

# Visualization: Scatterplot with Regression Line
ggplot(project_data, aes(x=VaccinationRate, y=DeathRate)) +
  geom_point(alpha=0.4, color="blue") +
  geom_smooth(method="lm", color="red", se=TRUE) +
  labs(title="Linear Regression: Vaccination vs Death Rate", x="Vaccination Rate (%)",
y="Death Rate") +
  theme_minimal()

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# PART V: Statistical Inference (Confidence Intervals)
# (Population Mean Estimation)
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cat("\n--- PART V: CONFIDENCE INTERVALS ---\n")

# Calculate 95% Confidence Interval for the Mean Death Rate
ci_test <- t.test(project_data$DeathRate, conf.level = 0.95)
print(ci_test)
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# Interpretation Helper
cat(sprintf("We are 95%% confident that the true population mean of daily death rates falls between %.3f and %.3f.\n",
        ci_test$conf.int[1], ci_test$conf.int[2]))

# End of Code
```